

Continuación de la clase anterior

rec: P 1.8 $\dot{g} = k_1 s_0 - k_2 g + \frac{k_3 g^2}{k_4^2 + g^2} \xrightarrow{\text{lo convertimos en}} \frac{dx}{dz} = s - rx + \frac{x^2}{1+x^2}$

PF: para $s=0$ (dejo de poner reactante al experimento y veo qué hace el reactivo)

$$-rx + \frac{x^2}{1+x^2} = 0 \Leftrightarrow x \left(\frac{x}{1+x^2} - r \right) = 0$$

$$x^* = 0, x^* =$$

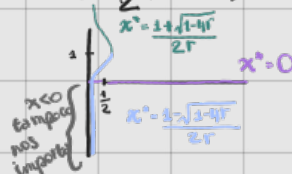
$$\odot x_{1,2}^*(r=r_c) = 1 \neq x_0 \quad \text{"Parece nodo silla"}$$

↗ o sea no nacen en $x=0$

⇒ no es algo de la forma $\left(\begin{array}{c} \text{---} \end{array} \right)$

$$x = r + x^2 r \rightarrow x_{1,2}^* = \frac{1 \pm \sqrt{1-4r^2}}{2r}$$

$$\exists \text{ si } 1-4r^2 \geq 0 \Leftrightarrow \frac{1}{2} \geq r \geq 0 \Rightarrow r_c = \frac{1}{2}$$



Vemos estabilidad:

$$\frac{df}{dx} = -r + \frac{2x}{(1+x^2)^2} \Rightarrow \frac{df}{dx} \Big|_0 = -r < 0 \quad \text{estable siempre}$$

pues $r < 0$ no tiene sentido físico

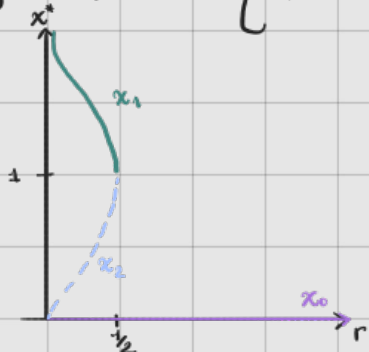
Para $x_{1,2}$ uso $r = \frac{x_{1,2}}{1+x_{1,2}^2} \Rightarrow \frac{df}{dx} \Big|_{x_{1,2}} = \frac{-x_{1,2}}{(1+x_{1,2}^2)} + \frac{2x_{1,2}}{(1+x_{1,2}^2)^2} = \frac{x_{1,2} - x_{1,2}^3}{(1+x_{1,2}^2)^2} = \frac{x_{1,2}(1-x_{1,2}^2)}{(1+x_{1,2}^2)^2}$

(eliminamos solo $x_{1,2} > 0$)

> 0

$$\Rightarrow \text{sg} \left(\frac{df}{dx} \Big|_{x_{1,2}} \right) = \text{sg}(1-x_{1,2}^2) = \pm 1 \Rightarrow \begin{cases} x_2 < 1 \text{ es inestable} \\ x_1 > 1 \text{ es estable} \end{cases}$$

Nos queda



Comentario: ¿y si $s \neq 0$?

Sup $x(0)=0, s(t) \nearrow$ (va aumentando lentamente)

Nuevos PF: $S(x) = rx - \frac{x^2}{1+x^2}$, a $t=T$ se apaga s

→ Si $x(T) < x_2$ me voy a $x=0$

→ Si $x(T) > x_2$ me voy a $x=x_1$

Sistemas Lineales 2D

$$\begin{aligned} \dot{x} &= ax + by \\ \dot{y} &= cx + dy \end{aligned} \Rightarrow \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \equiv \dot{\bar{x}} = A \bar{x}$$

PF: $A \bar{x}^* = 0$

- A invertible ($\det(A) \neq 0$) $\Rightarrow \bar{x}^* = 0$ es el único PF (se nota "aislado")
- A no invertible ($\det(A) = 0$) $\Rightarrow \exists$ recta de PF.

Busco $\bar{x}(t) = e^{\lambda t} \bar{v} \Rightarrow \lambda e^{\lambda t} \bar{v} = A e^{\lambda t} \bar{v} \leftarrow \text{O sea buscamos autovectores } \lambda \bar{v} = A \bar{v}$

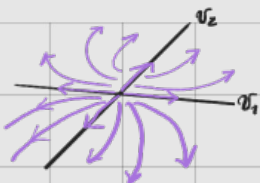
Posibilidades

- $\lambda_1, \lambda_2 \in \mathbb{R}$ (ambos)
- $\lambda_1, \lambda_2 \in \mathbb{C}$ con $\bar{\lambda}_1 = \lambda_2$
- $\lambda_1 = \lambda_2 \in \mathbb{R}$

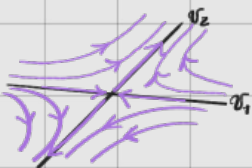
Caso 1: autovalores distintos y reales

Autovalores $\lambda_1, \lambda_2 \in \mathbb{R}$ con autovectores $\bar{v}_1, \bar{v}_2 \Rightarrow \bar{x}(t) = c_1 e^{\lambda_1 t} \bar{v}_1 + c_2 e^{\lambda_2 t} \bar{v}_2$

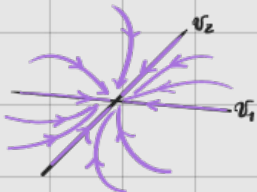
Fuente (nodo repulsor) $0 < \lambda_1 < \lambda_2$



Punto silla $\lambda_1 < 0 < \lambda_2$



Nodo atractor $\lambda_1 < \lambda_2 < 0$



Sea $A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$

$\det(A - \lambda I)$

$\begin{vmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{vmatrix}$

$= (3-\lambda)(2-\lambda) - 2 \stackrel{\text{igual a 0 para obtener } \lambda_1, \lambda_2}{=} 0$

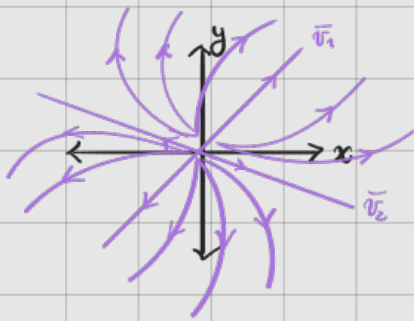
$\Leftrightarrow \lambda^2 - 5\lambda + 4 = 0 \Leftrightarrow \lambda_1 = 4, \lambda_2 = 1$ ($\lambda_1 > \lambda_2 > 0$ fuente (nodo repulsor))

Buscamos $\bar{v}_1, \bar{v}_2$

$\bar{v}_1: \begin{pmatrix} 3-\lambda_1 & 1 \\ 2 & 2-\lambda_1 \end{pmatrix} \begin{pmatrix} v_{1x} \\ v_{1y} \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} v_{1x} \\ v_{1y} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow v_{1x} = v_{1y} \Rightarrow \bar{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

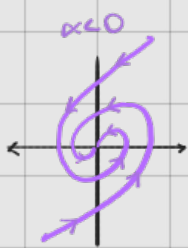
$\bar{v}_2: \begin{pmatrix} 3-\lambda_2 & 1 \\ 2 & 2-\lambda_2 \end{pmatrix} \begin{pmatrix} v_{2x} \\ v_{2y} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} v_{2x} \\ v_{2y} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow 2v_{2x} + v_{2y} = 0 \Leftrightarrow v_{2y} = -2v_{2x} \Rightarrow \bar{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

Luego $\bar{x}(t) = c_1 e^{4t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^t \begin{pmatrix} 1 \\ -2 \end{pmatrix} \Rightarrow \begin{cases} x(t) = c_1 e^{4t} + c_2 e^t \\ y(t) = c_1 e^{4t} - 2c_2 e^t \end{cases}$



Caso 2: autovalores complejos (conjugados)

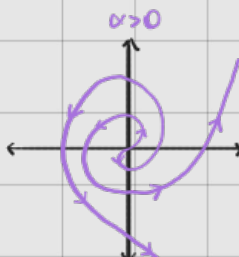
$\lambda = \alpha \pm i\beta, \bar{v} = \bar{v}_R + i\bar{v}_I, \bar{x}(t) = e^{\alpha t} \{ c_1 \operatorname{Re}(e^{i\beta t} \bar{v}) + c_2 \operatorname{Im}(e^{i\beta t} \bar{v}) \}$



Espiral/foco atractor



Centros



Espiral/foco repulsor

Ejemplo $\begin{cases} \dot{x} = 3x + 2y \\ \dot{y} = -4x - y \end{cases} \Rightarrow A = \begin{pmatrix} 3 & 2 \\ -4 & -1 \end{pmatrix} \Rightarrow \lambda_{1,2} = 1 \pm 2i$ (foco repulsor)

autovector: $\lambda_1 = 1 + 2i$

$(A - \lambda_1 I)(\bar{v}_1) = \vec{0} \Leftrightarrow \begin{pmatrix} 2-2i & 2 \\ -4 & -2-2i \end{pmatrix} \begin{pmatrix} v_{1x} \\ v_{1y} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow (2-2i)v_{1x} + 2v_{1y} = 0$ etc etc

$\bar{v}_1 = \begin{pmatrix} -1 \\ 1-i \end{pmatrix}$

$e^{i\beta t} \bar{v}_1 = \begin{pmatrix} -\cos(2t) \\ -\sin(2t) \end{pmatrix}$

